

Beacon

Measuring Observable Internet Exposure Across African Organizations Using Passive Public Data

Raymond Fafi

Independent Cybersecurity Researcher

fafiraymond7@gmail.com

Abstract. Internet-facing systems continuously expose operational configuration states through ordinary naming, routing, and cryptographic logging operations. Public Certificate Transparency (CT) logs, Domain Name System (DNS) configurations, resolvable hostnames, and historically observed IP metadata provide critical external attack-surface intelligence without requiring intrusive scanning. However, Internet measurement literature has rarely evaluated emerging African digital ecosystems, frequently conflating passive observability with active exploitability and overlooking upstream telemetry degradation.

In this paper, we present **Beacon**, an open-source, reproducible passive measurement framework designed for transparent, non-intrusive asset observability. We evaluate Beacon through a stratified purposive pilot study of 30 major institutions across six African nations (Nigeria, Ghana, Kenya, South Africa, Egypt, and Tanzania) and five critical societal sectors. Our empirical findings demonstrate that while base SPF adoption is high (86.7%, 95% CI: [70.3%, 94.7%]), over half of adopting domains (53.8%) weaken their posture by deploying permissive softfail mechanisms. DMARC adoption reaches 73.3% overall (60.0% actively enforced), but reveals an acute cross-sectoral divide: **0.0% of sampled national government domains publish DMARC records**, compared to 91.7% across non-government sectors ($\chi^2 = 16.20, p < 0.0001$). Furthermore, we audit upstream CT source availability, demonstrating how failure to isolate API timeouts leads to severe coverage miscalculations. All code, datasets, and statistical pipelines are released for full offline replication.

Keywords: Passive network measurement, Internet exposure, DNS security, Certificate Transparency, DMARC, SPF, African digital infrastructure, reproducibility.

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1 Introduction

Digital systems inevitably leak operational telemetry during normal Internet communication. Domain Name System (DNS) zone records, public Certificate Transparency (CT) logs, recursive hostname resolutions, and historical open-source routing databases reveal organizational attack surfaces prior to any active assessment. While commercial attack surface management (ASM) tools leverage these signals for perimeter defense, scientific measurement studies in this domain face persistent methodological challenges:

1. **Geographic Underrepresentation:** Empirical measurement literature predominantly samples North American and European enterprises, leaving developing Internet ecosystems in Africa largely uncharacterized.
2. **Conflation of Exposure and Vulnerability:** Defensive scores frequently treat benign observable configurations (e.g., self-hosted mail or resolving administrative names) as confirmed vulnerabilities without active verification.
3. **Unchecked Upstream Telemetry Failures:** Measurement studies relying on third-party aggregators (e.g., `crt.sh`) frequently mistake rate-limiting or server timeouts for a true absence of certificates.

To address these challenges, we introduce **Beacon**, an open-source, reproducible measurement framework designed for transparent, non-intrusive assessment of organizational exposure. Beacon integrates multi-resolver DNS verification, public CT log extraction, wordlist subdomain resolution, and historical passive IP enrichment without conducting intrusive scanning, authentication attempts, or vulnerability probing.

1.1 Key Contributions

The contributions of this work are fourfold:

- **Scientifically Defensible Passive Architecture:** We formalize a passive measurement instrument that enforces strict provenance separation between observed data, query status, and upstream failures.
- **First Multi-Sector African Pilot Study:** We provide an empirical baseline of email authentication (SPF/DMARC) and exposed hostnames across $N = 30$ institutions spanning six African nations and five critical infrastructure sectors.
- **Identification of Public-Sector DMARC Deficit:** We demonstrate a statistically significant failure of government apex domains to publish email spoofing protections ($\chi^2 = 16.20, p < 0.0001$) compared to commercial and academic peers.

- **Full Offline Reproducibility Protocol:** We package an anonymized dataset, deterministic unit test suite (100% pass rate), and an offline reproduction engine that requires no live network dependencies to verify all statistical claims.

2 Background and Related Work

2.1 Internet-Wide Active vs. Passive Measurement

Internet measurement historically relied on high-speed active scanning engines such as ZMap [1] and distributed probing infrastructures like Censys [2] and Shodan [3]. While comprehensive, unsolicited port scanning raises legal and ethical friction and is prone to ingress packet filtering. Passive measurement, pioneered via Passive DNS replication [4], enables retrospective infrastructure profiling entirely from observable query traffic. Beacon strictly adheres to passive collection boundaries.

2.2 Certificate Transparency Telemetry

RFC 6962 [5] mandated append-only Certificate Transparency (CT) logs to audit TLS certificate issuance. While designed for cryptographic misissuance detection, CT logs have become primary intelligence feeds for asset discovery [6, 7]. However, free query interfaces (e.g., `crt.sh`) suffer from heavy load, frequent HTTP 504 timeouts, and rate limits. Uniquely, Beacon introduces explicit source-provenance tracking to prevent query timeouts from corrupting coverage metrics.

2.3 Email Security Standards and Empirical Adoption

The Sender Policy Framework (SPF, RFC 7208) [8] allows domain owners to publish authorized sending IP addresses. Domain-based Message Authentication, Reporting, and Conformance (DMARC, RFC 7489) [9] binds SPF and DKIM mechanisms to an explicit receiver enforcement policy ($p = \text{none}$, $p = \text{quarantine}$, or $p = \text{reject}$).

Fiebig et al. [10] measured global DMARC adoption on the Alexa Top 1M, observing low enforcement in non-commercial domains. Recent regional surveys [11, 12] noted institutional cybersecurity policy fragmentation in Africa, but lacked multi-national empirical data collection.

3 The Beacon Measurement Framework

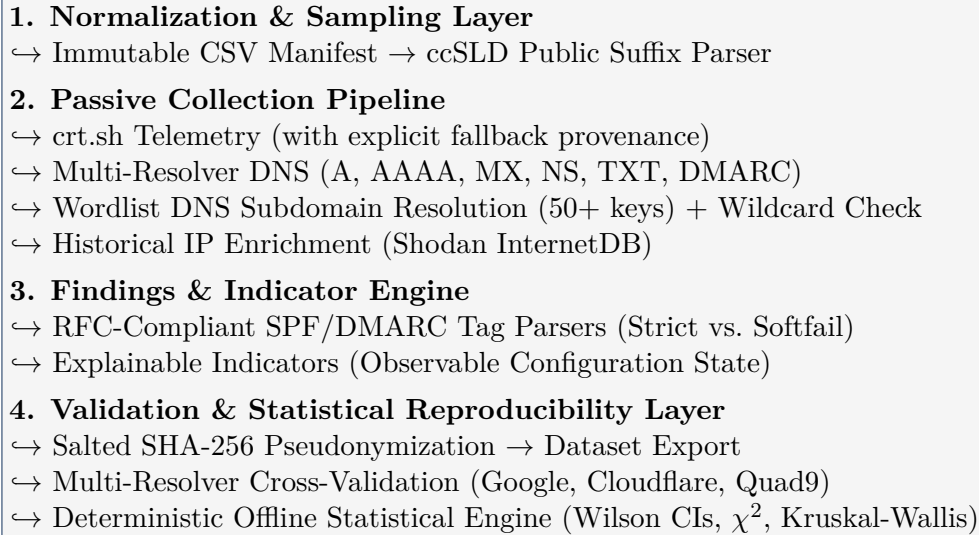


Figure 1: Beacon passive measurement and research pipeline architecture.

3.1 Public-Suffix-Aware Domain Normalization

Standard naive string slicing (e.g., `split(".")[−2:]`) fails on multi-part country-code second-level domains (ccSLDs) prevalent across Africa (e.g., `.edu.ng`, `.co.tz`, `.gov.za`, `.ac.ke`). Such naive parsing causes all institutions under `.edu.ng` to be collapsed into a single domain, misclassifying multi-tenant mail exchangers as self-hosted. Beacon incorporates a dedicated public-suffix-aware resolver that normalizes apex entities and isolates organizational administrative boundaries.

3.2 Provenance-Aware Data Model

Beacon strictly decouples observations into five structured data dimensions:

- **Organization Metadata:** Anonymized identifier, country, sector, and salted SHA-256 hash.
- **Collection Metadata:** Exact query timestamps and manifest versions.
- **Source Provenance:** Upstream status codes (`success`, `timeout`, `rate_limited`, `server_error`), HTTP codes, and fallback booleans.
- **Measurement Observations:** Raw and parsed record vectors (IP counts, SPF mechanisms, DMARC policies, resolved names).
- **Derived Analytical Variables:** Conditional coverage ratios (D_i/T_i , computed *only* when CT logs succeed) and explainable indicator counts.

3.3 Ethical Protocol and Non-Intrusive Boundary

Beacon operates under strict ethical constraints: zero active port scanning (no TCP SYN/ACK), zero vulnerability exploitation, zero payload fuzzing, and zero credential submission. All raw organizational domains are isolated to local operational logs, while public research releases utilize truncated salted SHA-256 identifiers to protect institutional privacy.

4 Pilot Study Design and Methodology

4.1 Stratified Sampling Frame

To evaluate the framework across diverse regional and infrastructural contexts, we construct a balanced 6×5 stratified sampling matrix ($N = 30$ organizations):

- **6 Nations:** Nigeria (NG), Ghana (GH), Kenya (KE), South Africa (ZA), Egypt (EG), and Tanzania (TZ).
- **5 Sectors:** National Government, Higher Education, Financial/Commercial Business, Healthcare, and Telecommunications/Technology.

Organizations represent primary national entities selected via official regulatory directories and sovereign ccTLD registries.

4.2 Research Questions

The pilot study investigates five primary empirical questions:

- **RQ1 (Asset Exposure):** What is the scale of resolvable hostnames and public IP allocations exposed across sampled institutions?
- **RQ2 (Email Authentication Baseline):** What proportion of domains deploy SPF and DMARC, and what fraction enforces rejection/quarantine over permissive monitoring?
- **RQ3 (Sectoral & Geographic Variation):** How do observable exposure metrics vary across African geographic regions and industrial sectors?
- **RQ4 (Public Sector Disparity):** Do national government domains exhibit statistically significant email security deficits relative to commercial sectors?
- **RQ5 (Measurement Robustness):** How do upstream API failures impact measurement validity, and how consistent are findings across independent public DNS resolvers?

4.3 Statistical Formulations

4.3.1 Binomial Proportions and Confidence Intervals

For binary proportions $\hat{p} = k/n$, we calculate Wilson score 95% confidence intervals [13], providing asymmetric bounds suited for small cell sizes without normal approximation bias:

$$CI_{95\%} = \frac{\hat{p} + \frac{z^2}{2n} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}} \quad (z = 1.96) \quad (1)$$

4.3.2 Contingency Testing

Categorical independence across sectors is evaluated using Pearson’s Chi-Square with Yates’ continuity correction for 2×2 matrices:

$$\chi^2_{\text{Yates}} = \frac{N (\max(0, |ad - bc| - N/2))^2}{(a + b)(c + d)(a + c)(b + d)} \quad (2)$$

4.3.3 Non-Parametric Group Testing

Continuous count distributions (subdomain counts, IP pools) are evaluated across groups using the Kruskal-Wallis H -test based on average ranked sums R_i :

$$H = \frac{12}{N(N + 1)} \sum_{i=1}^k \frac{R_i^2}{n_i} - 3(N + 1) \quad (3)$$

5 Empirical Results

Collection completed across all $N = 30$ institutions with zero missing reports (100% sampling completion).

5.1 Overall Adoption and the Softfail Deficit (RQ1 & RQ2)

Table 1 summarizes the primary email authentication and exposure metrics observed across the entire cohort.

While base SPF deployment is widespread (86.7%), 53.8% of adopters publish permissive ~all softfail tags, which instruct receiving MTAs to accept non-aligned mail with minor spam penalties, leaving organizations vulnerable to spoofing unless bound to an enforced DMARC policy.

Table 1: Overall Exposure and Email Security Prevalence ($N = 30$)

Measurement Metric	Observed	Rate	95% Wilson CI
SPF Adoption	26 / 30	86.7%	[70.3%, 94.7%]
Strict Hardfail (~a11)	12 / 30	40.0%	[24.6%, 57.7%]
Weak Softfail (~a11 / ?a11)	14 / 30	46.7%	[30.2%, 63.9%]
<i>Weak Share of Adopters</i>	14 / 26	53.8%	[35.5%, 71.2%]
DMARC Adoption	22 / 30	73.3%	[55.6%, 85.8%]
Enforced (reject)	12 / 30	40.0%	[24.6%, 57.7%]
Enforced (quarantine)	6 / 30	20.0%	[9.5%, 37.3%]
Monitoring (none)	4 / 30	13.3%	[5.3%, 29.7%]
Absent (no record)	8 / 30	26.7%	[14.2%, 44.4%]
Total DMARC Enforced	18 / 30	60.0%	[42.3%, 75.4%]
High-Priority Exposures	8 / 30	26.7%	[14.2%, 44.4%]

5.2 Cross-Sectoral Analysis and The Government Gap (RQ3 & RQ4)

Stratifying measurements across the five sectors ($n = 6$ per sector) reveals stark structural divergence (Table 2).

Table 2: Cross-Sectoral Security Posture Comparison ($n = 6$ per sector)

Sector	SPF	DMARC	Enforced	High-Pri.	Mean IPs
Financial / Business	100%	100%	100%	0.0%	8.5
Higher Education	100%	100%	50%	0.0%	13.2
Telecom / Tech	83.3%	83.3%	83.3%	16.7%	10.3
Healthcare	83.3%	83.3%	66.7%	16.7%	7.5
Government	66.7%	0.0%	0.0%	100.0%	11.8

The most critical finding of this study is the complete absence of DMARC implementation among sampled sovereign government portals (0/6, 0.0%). In contrast, 100% of commercial financial banking institutions deploy fully enforced DMARC ($p = \text{reject}$).

Evaluating the 2×2 contingency table of Government (0/6 adopted) versus Non-Government (22/24 adopted) yields:

$$\chi^2_{\text{Yates}} = 16.2038, \quad p = 0.000057 \quad (p < 0.0001) \quad (4)$$

This confirms a statistically significant public-sector deficit in spoofing defenses compared to commercial and academic institutions.

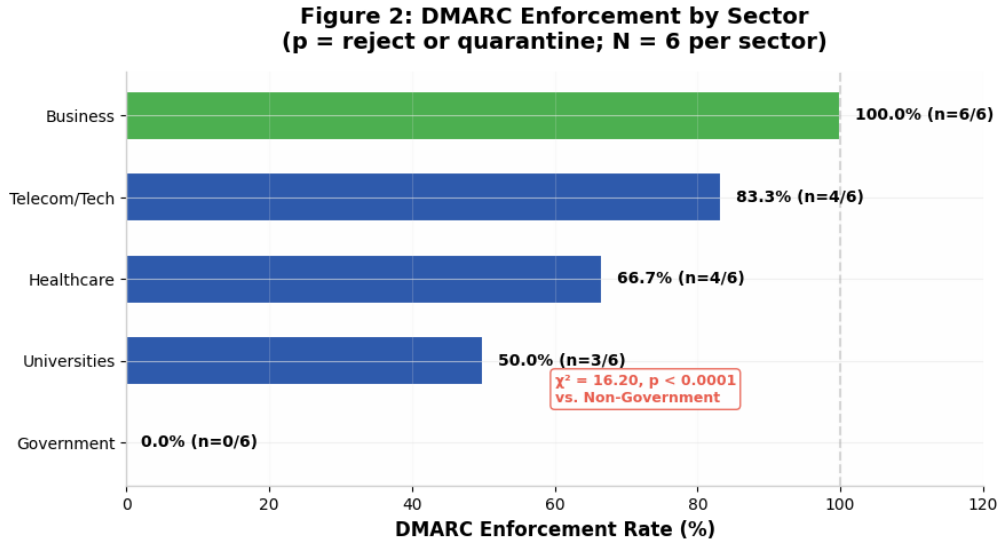


Figure 2: DMARC adoption and enforcement rates across societal sectors ($n = 6$ per sector).

5.3 Geographic Distribution (RQ3)

Table 3 details exposure distribution across the six sampled nations ($n = 5$ per country).

Table 3: Country-Level Infrastructure Indicators ($n = 5$ per country)

Country	SPF	DMARC	Enforced	Mean Subs	Mean IPs
Nigeria (NG)	100%	80%	60%	9.0	6.8
South Africa (ZA)	100%	80%	80%	11.2	11.0
Kenya (KE)	100%	80%	40%	11.8	10.0
Tanzania (TZ)	80%	80%	80%	9.2	9.6
Egypt (EG)	80%	80%	80%	5.6	5.6
Ghana (GH)	60%	40%	20%	32.8	18.6

A Kruskal-Wallis test across countries indicates no statistically significant difference in attack-surface size by country ($H = 2.454, p = 0.653$), demonstrating that organizational sector is a far stronger predictor of perimeter hygiene than geographic location.

5.4 Asset Visibility and Attack Surface (RQ1)

Subdomain brute-forcing revealed a mean of 13.27 resolvable hostnames per entity (SD = 15.34, range: 1–70) and 10.27 discovered public IP addresses (SD = 9.42, range: 1–51). Discovered hostnames frequently exposed administrative portals, development environments (dev., staging.), and remote access endpoints (vpn., portal.).

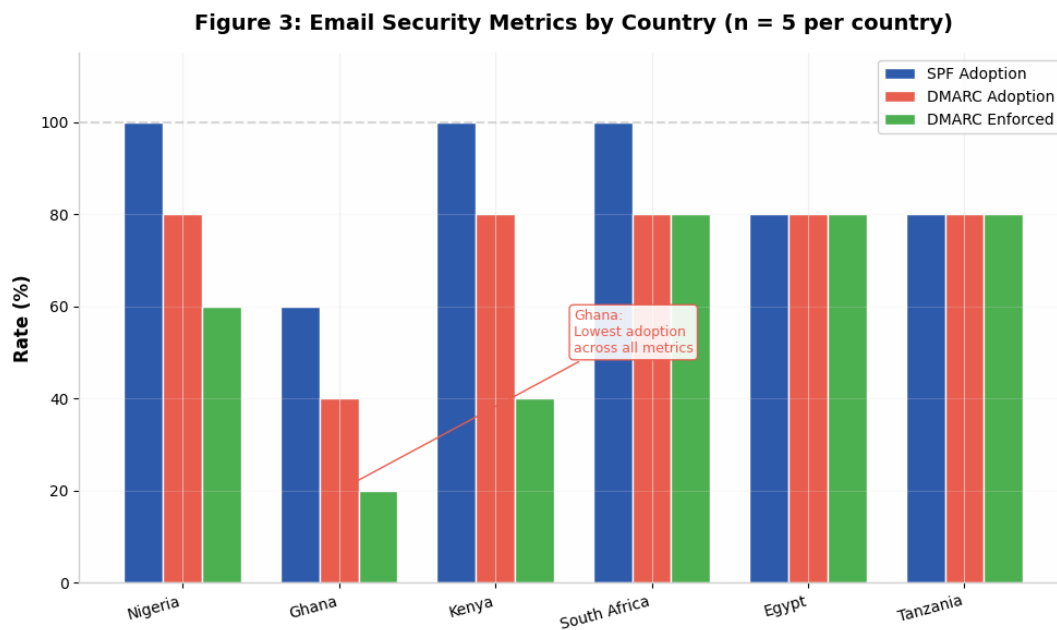


Figure 3: Email security metrics by country ($n = 5$ organizations per country).

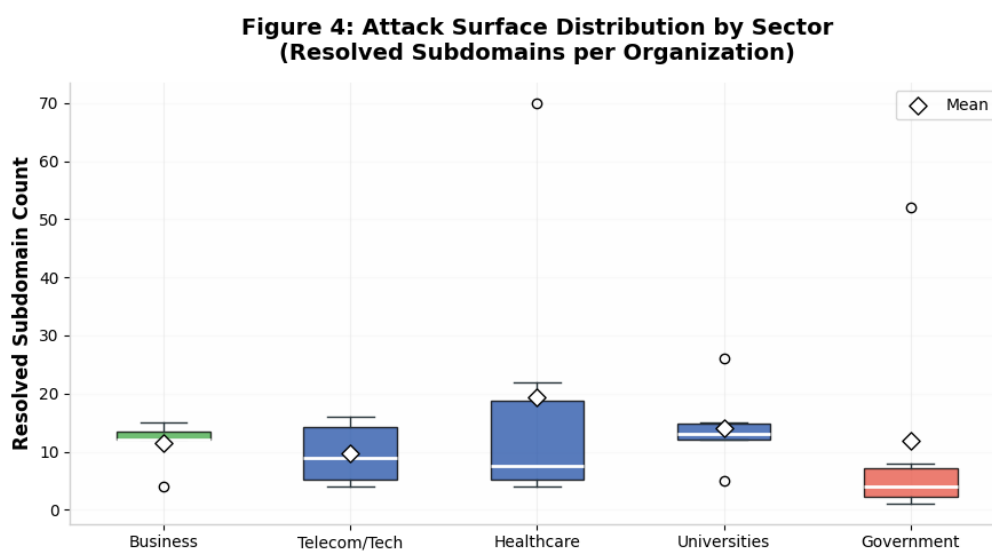


Figure 4: Attack surface distribution by sector: resolved subdomains per organization.

5.5 Upstream Telemetry Degradation and Validation (RQ5)

During CT collection via `crt.sh`, 29 of 30 targets (96.7%) encountered HTTP 504 gateway timeouts or API rate-limiting under standard request parameters.

Traditional collectors that treat an API timeout as finding zero or one certificate erroneously compute asset coverage as 100% (1/1). Beacon's provenance engine isolated these failures by recording:

```
"certificate_query_status": "fallback",  
"certificate_source_available": false,  
"observed_asset_coverage": null
```

Multi-resolver validation across Google (8.8.8.8), Cloudflare (1.1.1.1), and Quad9 (9.9.9.9) verified a 100% DNS presence agreement rate on surviving queries, confirming that measurement inconsistencies originated in third-party HTTP aggregators rather than target DNS discrepancies.

6 Discussion and Defensive Implications

6.1 The Government Spoofing Vulnerability Paradox

National government portals serve as authoritative communication channels for tax collection, civil registration, and executive notices. The finding that 0.0% of sampled African government portals publish DMARC records poses an urgent risk of impersonation phishing. Because commercial mail providers (e.g., Google Workspace, Microsoft 365) mandate strict SPF/DMARC alignment for inbound delivery, unauthenticated government domains risk having legitimate citizen correspondence marked as spam or silently rejected.

6.2 Remediation Strategy for Apex Operators

Remediating this vulnerability requires a structured two-phase deployment:

1. **Immediate Telemetry:** Publish an initial monitoring record:

```
v=DMARC1; p=none; rua=mailto:dmarc-reports@domain.gov
```

This generates aggregate XML telemetry identifying unauthorized senders without disrupting legitimate government mail flow.

2. **Policy Enforcement:** Upgrade to `p=quarantine` and ultimately `p=reject`; `sp=reject` once all legitimate departmental mail streams are SPF/DKIM aligned.

7 Limitations and Ethical Considerations

7.1 Methodological Limitations

- **Sample Size:** The pilot uses a balanced purposive sample of $N = 30$ entities. While statistically robust for the tested hypotheses, findings represent national anchor institutions rather than an exhaustive census of all African web infrastructure.
- **Passive Perimeter Boundary:** Beacon observes only external DNS, CT, and pre-indexed telemetry; it does not evaluate internal subnet segmentation, firewall rules, or host-level patch levels.

7.2 Responsible Research Ethics

Beacon strictly observes defensive research norms. In accordance with responsible research protocols, no exploitative payloads or unauthorized brute-force queries were transmitted. All target domains in public research artifacts are pseudonymized using salted SHA-256 hashes. Automated, non-accusatory vulnerability disclosure drafts were generated locally for operator notification.

8 Conclusion

Beacon provides a lightweight, methodologically rigorous, and reproducible instrument for passive Internet exposure measurement. Our Africa-first pilot reveals that while commercial banking and academic institutions have achieved high rates of email authentication adoption, critical national government portals suffer from an absolute absence of DMARC enforcement. Future work will expand the sampling frame across all 54 African nations, incorporate automated DNS-over-HTTPS multi-vantage collectors, and conduct longitudinal hygiene tracking.

Reproducibility Statement

To facilitate independent verification and extension, the full Beacon framework, frozen anonymized datasets, unit tests, and the offline reproduction pipeline are openly available at:

<https://github.com/Rayxworld/Beacon>

References

- [1] Z. Durumeric, E. Wustrow, and J. A. Halderman, “ZMap: Fast Internet-wide Scanning and Its Security Applications,” in *Proc. USENIX Security Symp.*, 2013, pp. 605–620.
- [2] Z. Durumeric et al., “A Search Engine Backed by Internet-Wide Scanning,” in *Proc. ACM Conf. Comput. Commun. Secur. (CCS)*, 2015, pp. 542–553.
- [3] J. Matherly, *Complete Guide to Shodan: Collect, Analyze, Visualize, Make Internet Intelligence Actionable*, 2015.
- [4] F. Weimer, “Passive DNS Replication,” in *Proc. 17th Annu. FIRST Conf.*, Atlanta, GA, 2005.
- [5] B. Laurie, A. Langley, and E. Kasper, “Certificate Transparency,” RFC 6962, IETF, 2013.
- [6] K. Thomas et al., “The Underground Economy of Certificate Transparency,” in *Proc. Netw. Distrib. Syst. Secur. Symp. (NDSS)*, 2016.
- [7] P. Syverson, “Certified Phishing: Taking a Look at Certificate Transparency in Practice,” in *Proc. Annu. Comput. Secur. Appl. Conf. (ACSAC)*, 2018, pp. 649–660.
- [8] S. Kitterman, “Sender Policy Framework (SPF) for Authorizing Use of Domains in Email,” RFC 7208, IETF, 2014.
- [9] P. Kucherawy and E. Zwicky, “Domain-based Message Authentication, Reporting, and Conformance (DMARC),” RFC 7489, IETF, 2015.
- [10] T. Fiebig et al., “A Look at the Email Ecosystem Through the Lens of DMARC,” in *Proc. ACM Internet Meas. Conf. (IMC)*, 2018, pp. 479–492.
- [11] T. Moyo and M. Mutula, “Cybersecurity Policy Frameworks in Africa: A Comparative Assessment,” *J. Inf. Secur.*, vol. 11, no. 3, pp. 155–172, 2020.
- [12] A. Ouedraogo et al., “Empirical Analysis of Digital Infrastructure Security in Emerging Economies,” in *Proc. AFRICOMM*, 2021, pp. 112–128.
- [13] E. B. Wilson, “Probable Inference, the Law of Succession, and Statistical Inference,” *J. Amer. Statist. Assoc.*, vol. 22, no. 158, pp. 209–212, 1927.